

Global and Per-Sector Unit-Weight Witnesses in the CPSV16 BNT Argument

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1 The source normalization

Cirac–Perez-Garcia–Schuch–Verstraete 2016 normalizes the raw coefficients in the basis-of-normal-tensors (BNT) expansion by a single global condition. Section II.C, line 246 states the condition for the one-index canonical-form coefficients μ_k . The BNT display at lines 287–289 then refines this index into a sector index and a copy index. In the resulting two-layer notation

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1},$$

the source assumes

$$|\mu_{j,q}| \leq 1 \quad \text{for all } (j,q), \quad |\mu_{j_*,q_*}| = 1 \quad \text{for some } (j_*,q_*).$$

This is the same normalization, expressed after the two-layer BNT refinement.

Thus the paper-level canonical form does not state that every BNT sector j contains a copy q of modulus one. A sector may contain only strictly subunit copy weights, while some other sector carries the global unit witness.

2 The formal boundary

The formal predicate `MPSTensor.IsBNTCanonicalForm` records the source normalization faithfully: it has a bound on all copy weights and a single global unit-weight witness. It does not contain the per-sector condition

$$\forall j, \exists q, |\mu_{j,q}| = 1.$$

Several full-basis matching and global-gauge theorems currently assume this per-sector condition explicitly. The condition is used to apply the Cesaro non-decay lemma sector by sector in the CPSV16 Appendix MPV line-1182 matching argument. The affected formal argument includes the strong matching theorem,

the bijective matching theorem, the proportional matched-sector theorem, and the equal-MPV global-gauge theorems in the SectorBNT formulation.

Consequently these declarations are not the unrestricted CPSV16 theorem or corollary. They are scope-restricted full-basis statements: after the per-sector unit-weight hypotheses are supplied, they prove the sector matching, coefficient comparison, copy-weight recovery, and direct-sum gauge construction without imposing any strict ordering of the copy-weight moduli.

3 Equal-modulus declaration path

The current formal assembly is organized through `TNLean/MPS/FundamentalTheorem/SectorBNT/`. There is no separate `TNLean/MPS/FundamentalTheorem/Multi/` assembly directory on the current main branch.

The equal-modulus question is logically separate from the per-sector unit-witness restriction above. The current SectorBNT coefficient comparison works with raw sector coefficients

$$c_N^{(j)} = \sum_q \mu_{j,q}^N$$

and the copy weights are recovered from these coefficients by the finite power-sum argument. Equal-modulus copies such as the two copies in $C \oplus (-C)$, and unequal-modulus copies such as $C \oplus (1/2)C$, are both represented by the raw two-layer SectorBNT formulation.

The declaration path is as follows.

- `MPSTensor.IsBNTCanonicalForm` records the raw copy weights $\mu_{j,q}$, their coefficient sums, the bound $|\mu_{j,q}| \leq 1$, and the single global unit witness. It imposes neither a strict ordering of the moduli nor an equal-modulus layer.
- Sectorwise non-decay is supplied, after a unit-modulus copy in the given sector is available, by the declarations

`MPSTensor.CesaroNonDecay.sum`pow`not`tendsto`zero`of`unit`modulus`
`MPSTensor.IsBNTCanonicalForm.coeff`not`eventually`zero.`

- Eventual equality of the sector coefficient sequences is converted into equality of copy weights by `MPSTensor.matched`sector`weight`multiset`eq` and `MPSTensor.matched`sector`weight`equiv`, using the finite power-sum recovery theorem

`Matrix.sum`pow`eq`implies`card`eq`and`multiset`eq`of`le`max`card.`

- `MPSTensor.SectorBNTCopyWeightMatching.of`coeff`identity` packages this recovery into the copy-weight matching used by the proportional global-gauge theorem.

- In the equal-MPV path, the declarations

```

MPSTensor.ft'sector'bnt'equal'sector'data
MPSTensor.ft'sector'bnt'equal'global'gauge
MPSTensor.ft'sector'bnt'equal'mps'gaugeEquiv'literal

```

use the same coefficient-to-copy-weight step.

Thus equal-modulus examples such as $C \oplus (-C)$ are handled by the raw coefficient comparison and finite power-sum recovery after sector matching. No theorem in this SectorBNT declaration path assumes a strict ordering of the copy-weight moduli.

The remaining extra hypothesis is not strictness of moduli. It is the per-sector existence of a unit-modulus copy weight, which is stronger than the single global unit witness stated in CPSV16 line 246.

4 Elimination plan

There are two possible ways to remove the restriction.

First, one may prove the CPSV16 matching step directly from the global witness. In that approach, the line-1182 argument must show that every sector relevant to the full-basis matching either receives a unit-modulus witness from the source normalization after the appropriate comparison, or is eliminated by the same non-decay contradiction used in the paper.

Second, one may split the BNT basis into the unit-supporting part and the strictly subunit part, prove matching on the unit-supporting part, and then derive the remaining sectors by the coefficient identities. This argument must still end with the same raw coefficient comparison and finite power-sum recovery used by the current SectorBNT theorems.

Until one of these arguments is formalized, the per-sector statements should remain marked as scope-restricted versions of the BNT Fundamental Theorem argument. They should not be cited as the unrestricted CPSV16 theorem or corollary.

5 Current declarations

The main declarations carrying the additional per-sector hypothesis are:

- `MPSTensor.forall'k'exists'j'nondecaying'overlap'of'sameMPV,`
- `MPSTensor.bijjective'match'of'sameMPV,`
- `MPSTensor.ft'sector'bnt'proportional'sector'match'witnesses,`
- `MPSTensor.ft'sector'bnt'proportional'global'gauge'of'copy'weight'matching,`
- `MPSTensor.ft'sector'bnt'proportional'global'gauge'of'coeff'identity,`

- `MPSTensor.ft`sector`bnt`equal`global`gauge`,
- `MPSTensor.ft`sector`bnt`equal`mps`gaugeEquiv`literal`.

The faithful structural normalization remains the global unit-weight field `MPSTensor.IsBNTCanonicalForm.weight_unit_exists` in `TNLean/MPS/FundamentalTheorem/SectorBNT/Basic.lean`.

References